Gradient Descent and Optimization

ANSWER KEY

Section 1: Loss Function Derivatives and Stationary Points

1 Find the stationary point (global minimum) of the scalar mean squared error loss $L(w) = 3(w-4)^2 + 7$.

a. $w = 4$ ✓

b. $w = 7$

c. $w = 3$

d. $w = 0$

$L'(w) = 6(w-4) = 0 \Rightarrow w = 4$. $L''(4) = 6 > 0$, so $w = 4$ is the minimum.

2 For the loss function $L(w) = w^4 - 4w^2 + 6$, find all values of w where the gradient is zero.

a. $w = 0, \pm\sqrt{2}$ ✓

b. $w = \pm 2$

c. $w = 0, \pm 2$

d. $w = \pm 1$

$L'(w) = 4w^3 - 8w = 4w(w^2 - 2) = 0 \Rightarrow w = 0$ or $w = \pm\sqrt{2}$.

3 In optimization, a critical point has $L'(w) = 0$ and $L''(w) = -8$. What type of point is this?

a. **Local Maximum** ✓

b. Local Minimum

c. Saddle Point

d. Inflection Point

A negative second derivative implies concavity downwards, confirming a local maximum.

Section 2: Multivariate Gradients and Directional Derivatives

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4 Compute the gradient vector ∇L of $L(w_1, w_2) = 3w_1^2 + 5w_2^2 - 2w_1w_2$ evaluated at $(w_1, w_2) = (2, 1)$.

a. $[10, 6]^T$ ✓

b. $[12, 10]^T$

c. $[8, 8]^T$

d. $[14, 4]^T$

$\frac{\partial L}{\partial w_1} = 6w_1 - 2w_2 = 12 - 2 = 10$. $\frac{\partial L}{\partial w_2} = 10w_2 - 2w_1 = 10 - 4 = 6$.

5 In which direction does the negative gradient vector $-\nabla L(\mathbf{w})$ point?

a. **The direction of steepest decrease of the loss function** ✓

b. The direction of steepest increase of the loss function

c. Orthogonal to the level curves in the direction of highest loss

d. Along the tangent line of the contour

∇L points in the direction of greatest increase; its negation $-\nabla L$ points in the direction of steepest descent.

6 Compute the directional derivative of $f(x, y) = x^2y$ at $(1, 2)$ in the unit direction $\mathbf{u} = [0.8, 0.6]^T$.

a. 3.8 ✓

b. 4.2

c. 2.6

d. 5.0

$\nabla f = [2xy, x^2]^T = [4, 1]^T$. Directional derivative $= \nabla f \cdot \mathbf{u} = 4(0.8) + 1(0.6) = 3.2 + 0.6 = 3.8$.

Section 3: Gradient Descent Step Updates

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7 A model weight has value $w_0 = 5.0$. If $\frac{\partial L}{\partial w} = 4.0$ and learning rate is $\eta = 0.1$, find the updated weight w_1.

a. 4.6 ✓

b. 5.4

c. 4.0

d. 4.9

Update: $w_1 = w_0 - \eta \nabla L = 5.0 - 0.1(4.0) = 5.0 - 0.4 = 4.6$.

8 Consider the quadratic loss $L(w) = \frac{1}{2}aw^2$ with $a = 10$. What is the maximum learning rate η before gradient descent diverges?

a. $\eta < 0.2$ ✓

b. $\eta < 0.1$

c. $\eta < 0.05$

d. $\eta < 1.0$

Update is $w_{t+1} = w_t(1 - \eta a)$. Stability requires
$|1 - \eta a| < 1 \Rightarrow -1 < 1 - 10\eta < 1 \Rightarrow 0 < \eta < 2/10 = 0.2$.

9 Perform one vector gradient descent update from $\mathbf{w}_0 = [1, 2]^T$ with $\nabla L = [2, -4]^T$ and $\eta = 0.25$.

a. $[0.5, 3.0]^T$ ✓

b. $[1.5, 1.0]^T$

c. $[0.5, 1.0]^T$

d. $[1.0, 3.0]^T$

$\mathbf{w}_1 = [1, 2]^T - 0.25[2, -4]^T = [1 - 0.5, 2 - (-1)]^T = [0.5, 3.0]^T$.

Section 4: Momentum and Advanced Optimizers

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10 In Polyak momentum, the velocity update is $\mathbf{v}_{t+1} = \beta \mathbf{v}_t + \eta \nabla L$. If $\beta = 0.9$, $\mathbf{v}_0 = 2.0$, $\eta = 0.1$, and $\nabla L = 4.0$, find $\mathbf{v}_1$.

a. 2.2 ✓

b. 2.0

c. 1.8

d. 2.6

$\mathbf{v}_1 = 0.9(2.0) + 0.1(4.0) = 1.8 + 0.4 = 2.2$.

11 Why does the Adam optimizer maintain running exponential averages of both first moments (mean) and second raw moments (uncentered variance)?

a. **To adapt per-parameter learning rates inversely proportional to gradient magnitude** ✓

b. To guarantee convergence to the global minimum in non-convex losses

c. To replace backpropagation with finite differences

d. To eliminate the need for mini-batch sampling

The second moment scales steps down for parameters with large, frequent gradients and up for sparse gradients.

12 What is the condition on the Hessian matrix $\mathbf{H} = \nabla^2 L(\mathbf{w})$ for a loss function to be strictly convex everywhere?

a. **$\mathbf{H}$ is positive definite everywhere (all eigenvalues > 0)** ✓

b. $\mathbf{H}$ has determinant equal to zero

c. $\mathbf{H}$ has trace equal to 1

d. $\mathbf{H}$ is skew-symmetric

Strict convexity requires a positive definite Hessian: $\mathbf{v}^T \mathbf{H} \mathbf{v} > 0$ for all non-zero $\mathbf{v}$.

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